Arbitrage Geometry

A Mathematical Theory of Structured Misalignment

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Abstract

This paper proposes *Arbitrage Geometry*, a new branch of mathematics devoted to spaces equipped with multiple valuation systems, admissible transformations, and transformation costs. The central objects are not points, curves, groups, or manifolds alone, but *exploitable discrepancies*: local or global misalignments between value, cost, information, strategy, power, risk, and representation. The theory abstracts financial arbitrage, statistical arbitrage, political arbitrage, computational shortcuts, strategic asymmetry, geopolitical imbalance, and warfare advantage into a single mathematical language. We define arbitrage spaces, arbitrage cells, arbitrage curvature, arbitrage gradients, arbitrage tensors, no-arbitrage equilibria, and arbitrage homology. A no-arbitrage flatness theorem is proved: on a connected transformation graph, zero arbitrage around every closed loop is equivalent to the existence of a global potential representation. Applications are given to economics, finance, political science, geopolitics, international relations, warfare economics, statistics, machine learning, and artificial intelligence.

The paper ends with "The End"

Contents

1 Introduction

Many mathematical disciplines study consistency. Geometry studies invariant structure under transformation; topology studies persistence under deformation; algebra studies lawful composition; analysis studies limiting behavior; probability studies coherent uncertainty; optimization studies the best feasible choice under a fixed objective.

Modern systems, however, are often governed not by one coherent valuation, but by many partially incompatible valuations. Markets contain prices, beliefs, liquidity premia, risk premia, regulatory costs, and information asymmetries. Political systems contain legitimacy, coercive power, coalition value, institutional constraints, and narrative capital. Wars contain terrain, logistics, morale, intelligence, technology, and attrition. Statistical systems contain likelihood, loss, prior belief, predictive accuracy, sampling bias, and model uncertainty. Machine learning systems contain training loss, generalization error, architecture cost, representation compression, inference latency, and deployment risk.

The motivating claim of this paper is that these domains share a common mathematical object:

structured exploitable misalignment.

We call the proposed branch studying such objects *Arbitrage Geometry.*

Classical arbitrage in finance is a transaction yielding positive payoff with no net cost and no risk. Arbitrage Geometry generalizes this idea. An arbitrage opportunity is any admissible transformation whose valuation gain exceeds its transformation cost. The core inequality is

$$V_j(Tx) - V_i(x) - C(T) > 0,$$

where x is a state, T is an admissible transformation, V_i, V_j are valuation functionals, and $C(T)$ is the cost of transformation.

The theory therefore studies systems of the form

$$\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C),$$

where X is a state space, $\mathcal{T}$ is a transformation system, $\mathcal{V}$ is a family of valuations, and C is a cost functional.

2 Motivating Examples

2.1 Financial arbitrage

Let X be a set of portfolios, $V_i(x)$ the market value of portfolio x under venue i, and T a trade moving the portfolio from one venue to another. If

$$V_j(Tx) - V_i(x) > C(T),$$

then the trader has a positive arbitrage margin.

2.2 Statistical arbitrage

Let X be a class of statistical models. Let $V_i(P_\theta)$ denote predictive performance under loss L_i, and let $C(T)$ be the complexity penalty of changing representation. A model transformation T is statistically profitable if

$$\mathrm{E}[L_i(P_\theta)] - \mathrm{E}[L_j(TP_\theta)] > C(T).$$

2.3 Computational arbitrage

Let x be a computational problem, T a representation change, $V(Tx)$ the reduction in runtime or sample complexity, and $C(T)$ the encoding cost. Computational arbitrage occurs when

$$\mathrm{Gain}(T, x) - \mathrm{EncodingCost}(T) > 0.$$

2.4 Political and geopolitical arbitrage

Let x be a political configuration, $V_i(x)$ the value assigned by actor i, and T a diplomatic, institutional, military, or informational move. A state can exploit geopolitical arbitrage when

$$V_j(Tx) - V_i(x) > C_{\text{diplomatic}}(T) + C_{\text{military}}(T) + C_{\text{reputational}}(T).$$

2.5 Warfare arbitrage

Let x be a theater state and let T be a maneuver, deception, strike, cyber operation, or logistical reallocation. A warfare arbitrage exists when the operational advantage exceeds the total cost:

$$V_{\text{enemy}}(x) - V_{\text{enemy}}(Tx) - C_{\text{operation}}(T) > 0.$$

This formalizes the idea that one actor may convert terrain, information, timing, or asymmetry into strategic value.

3 Primitive Objects

Definition 1 (Arbitrage space)**.** *An* arbitrage space *is a quadruple*

$$\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C),$$

where

- X *is a set of states;*
- $\mathcal{T}$ *is a set of admissible transformations* $T : X \to X$*, possibly partial;*
- $\mathcal{V} = \{V_i : X \to \mathbb{K}\}_{i \in I}$ *is a family of valuation functionals;*
- $C : \mathcal{T} \to \mathbb{K}$ *is a cost functional.*

Here $\mathbb{K}$ *is usually* $\mathbb{R}$*, but may also be* $\mathbb{C}$*, an ordered field, a vector lattice, a cone-ordered vector space, or a semiring.*

Definition 2 (Arbitrage functional)**.** *Given* $x \in X$*,* $T \in \mathcal{T}$*, and* $i, j \in I$*, the* arbitrage functional *is*

$$\Pi(T, x; i, j) = V_j(Tx) - V_i(x) - C(T).$$

Definition 3 (Positive arbitrage)**.** *A tuple* $(T, x; i, j)$ *is a* positive arbitrage *if*

$$\Pi(T, x; i, j) > 0.$$

It is a zero arbitrage *if* $\Pi(T, x; i, j) = 0$*, and a* negative arbitrage *if* $\Pi(T, x; i, j) < 0$*.*

Definition 4 (Arbitrage cell)**.** *An* arbitrage cell *is a local configuration*

$$\alpha = (U, \mathcal{T}_U, \mathcal{V}_U, C_U),$$

where $U \subseteq X$*,* $\mathcal{T}_U \subseteq \mathcal{T}$*,* $\mathcal{V}_U \subseteq \mathcal{V}$*, and at least one admissible transformation produces positive arbitrage:*

$$\exists x \in U, \quad \exists T \in \mathcal{T}_U, \quad \exists i, j \in I \quad \textit{such that} \quad \Pi(T, x; i, j) > 0.$$

4 The Geometric Interpretation

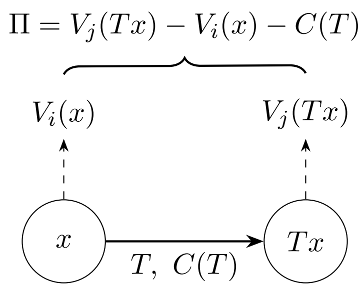
In ordinary geometry, the primitive question is often

$$d(x, y) = ?$$

In Arbitrage Geometry, the primitive question is

$$\Pi(T, x; i, j) = ?$$

Thus distance is replaced by discrepancy, displacement is replaced by transformation, and curvature is replaced by path-dependent accumulation of exploitable misalignment.



The diagram represents the fundamental object of the theory: an admissible movement from x to Tx, evaluated by two valuation systems and adjusted by cost.

5 Axioms of Arbitrage Geometry

Axiom 1 (State multiplicity)**.** *A system may admit more than one mathematically valid valuation:*

$$|\mathcal{V}| \geq 1.$$

Axiom 2 (Transformability)**.** *Value discrepancies become meaningful only relative to admissible transformations:*

$$T \in \mathcal{T}.$$

Axiom 3 (Cost obstruction)**.** *No transformation is free unless explicitly specified:*

$$C(T) \geq 0$$

when $\mathbb{K} = \mathbb{R}$.

Axiom 4 (Local exploitability)**.** *Positive arbitrage is a local property:*

$$\Pi(T, x; i, j) > 0.$$

Axiom 5 (Path dependence)**.** *The value of a sequence of transformations may depend on the path:*

$$\Pi(T_m \cdots T_1, x; i_0, i_m) \neq \sum_{k=1}^{m} \Pi(T_k, x_{k-1}; i_{k-1}, i_k).$$

When equality holds, the arbitrage space is said to be additively path-consistent along that path.

6 Arbitrage Curvature

Definition 5 (Transformation path)**.** *A transformation path is a sequence*

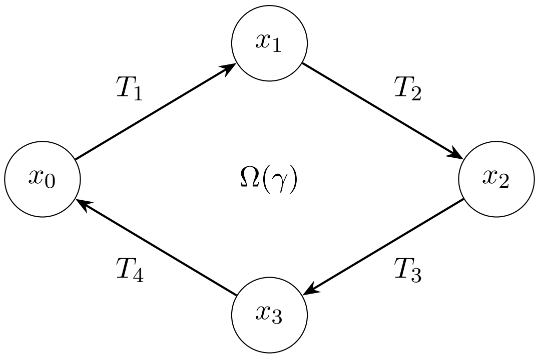
$$\gamma = (x_0, T_1, x_1, T_2, \ldots, T_m, x_m),$$

where $x_k = T_k x_{k-1}$*. It is closed if* $x_m = x_0$*.*

Definition 6 (Arbitrage curvature)**.** *Given a closed path* γ *and valuation indices* $i_0, \ldots, i_m$*, with* $i_m = i_0$*, define the arbitrage curvature by*

$$\Omega(\gamma) = \sum_{k=1}^{m} \left[V_{i_k}(x_k) - V_{i_{k-1}}(x_{k-1}) - C(T_k) \right].$$

If $\Omega(\gamma) > 0$, the closed path generates net arbitrage. If $\Omega(\gamma) = 0$, the loop is locally arbitrage-flat. If $\Omega(\gamma) < 0$, the loop dissipates value.



7 No-Arbitrage Equilibrium

Definition 7 (No-arbitrage equilibrium)**.** *A state $x^* \in X$ is a* no-arbitrage equilibrium *if*

$$\Pi(T, x^*; i, j) \leq 0$$

for all admissible $T \in \mathcal{T}$ and all valuation indices $i, j \in I$.

Definition 8 (Global no-arbitrage)**.** *The arbitrage space $\mathcal{A}$ is globally no-arbitrage if*

$$\Pi(T, x; i, j) \leq 0$$

for all $x \in X$, all $T \in \mathcal{T}$, and all $i, j \in I$.

Proposition 1 (Equilibrium under bounded gain)**.** *Suppose X is compact, $\mathcal{T}$ is finite, and each V_i is continuous. If*

$$\max_{x \in X, T \in \mathcal{T}, i, j \in I} \Pi(T, x; i, j) \leq 0,$$

then every $x \in X$ is a no-arbitrage equilibrium.

Proof. The assertion follows directly from the definition. Since the maximum of all arbitrage functionals is nonpositive, each individual arbitrage functional is nonpositive at every state. □

8 No-Arbitrage Flatness Theorem

We now give a foundational theorem. It is the analogue of the classical statement that a vector field with zero circulation on every closed path admits a scalar potential, provided the relevant domain is connected and path-independent.

Definition 9 (Transformation graph)**.** *Given an arbitrage space $\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C)$, its transformation graph is the directed graph*

$$\mathcal{G} = (\mathcal{N}, \mathcal{E}),$$

where a node is a valuation-state pair $(x, i) \in X \times I$, and an edge

$$e : (x, i) \to (y, j)$$

exists when there is $T \in \mathcal{T}$ such that $y = Tx$. The edge weight is

$$w(e) = V_j(y) - V_i(x) - C(T).$$

Theorem 1 (No-Arbitrage Flatness Theorem)**.** *Let $\mathcal{G}$ be a connected transformation graph. The following are equivalent:*

$$\sum_{e\in\gamma} w(e) = 0$$

for every closed directed cycle γ in $\mathcal{G}$, and there exists a potential function

$$P : X \times I \to \mathbb{K}$$

such that for every edge $e : a \to b$,

$$w(e) = P(b) - P(a).$$

Proof. First suppose there exists P such that $w(e) = P(b) - P(a)$ for every edge $e : a \to b$. For any closed cycle

$$\gamma : a_0 \to a_1 \to \cdots \to a_m = a_0,$$

we have

$$\sum_{k=1}^{m} w(a_{k-1}, a_k) = \sum_{k=1}^{m} [P(a_k) - P(a_{k-1})] = P(a_m) - P(a_0) = 0.$$

Conversely, suppose every closed cycle has zero total weight. Choose a base node $a_0 \in \mathcal{N}$. For any node a, choose a path $\rho : a_0 \to a$, and define

$$P(a) = \sum_{e\in\rho} w(e).$$

This is well-defined. Indeed, if ρ_1 and ρ_2 are two paths from a_0 to a, then following ρ_1 and then the reverse of ρ_2 gives a closed cycle. By hypothesis, its total weight is zero; hence both paths have the same total weight.

Now let $e : a \to b$ be any edge. Let ρ be a path from a_0 to a. Then $\rho + e$ is a path from a_0 to b, so

$$P(b) = P(a) + w(e).$$

Thus $w(e) = P(b) - P(a)$. □

Corollary 1 (Curvature as obstruction)**.** *Positive or negative arbitrage curvature around a closed loop is precisely the obstruction to global potential representation.*

9 Arbitrage Gradients and Tensors

Assume now that X is a differentiable manifold and each valuation $V_i : X \to \mathbb{R}$ is differentiable.

Definition 10 (Arbitrage gradient)**.** *For valuation pair (i, j), define the local arbitrage gradient by*

$$\nabla A_{ij}(x) = \nabla V_j(x) - \nabla V_i(x).$$

This vector field points in the local direction of steepest valuation discrepancy.

Definition 11 (Cost-adjusted arbitrage gradient)**.** *If $c_x(v)$ is the infinitesimal cost of moving from x in direction v, then the cost-adjusted arbitrage gain is*

$$G_{ij}(x, v) = \langle \nabla V_j(x) - \nabla V_i(x), v\rangle - c_x(v).$$

Definition 12 (Arbitrage tensor)**.** *Let $V : X \to \mathbb{R}^n$ be the vector valuation*

$$V(x) = (V_1(x), \ldots, V_n(x)) .$$

The arbitrage tensor is the matrix field

$$\mathsf{A}(x) = DV(x),$$

whose entries are

$$\mathsf{A}_{ik}(x) = \frac{\partial V_i}{\partial x_k}(x).$$

Pairwise arbitrage gradients are row differences of A.

10 Linear Arbitrage Geometry

Let $X = \mathbb{R}^d$, and suppose each valuation is linear:

$$V_i(x) = p_i^\top x,$$

where $p_i \in \mathbb{R}^d$. Let $T \in \mathbb{R}^{d\times d}$. Then

$$\Pi(T, x; i, j) = p_j^\top Tx - p_i^\top x - C(T).$$

Proposition 2 (Linear arbitrage condition)**.** *If $C(T) = c_T$ is constant in x, then positive arbitrage exists for some $x \in \mathbb{R}^d$ if and only if*

$$T^\top p_j - p_i \neq 0.$$

Proof. We have

$$\Pi(T, x; i, j) = (T^\top p_j - p_i)^\top x - c_T.$$

If $T^\top p_j - p_i = 0$, then $\Pi = -c_T \leq 0$ whenever $c_T \geq 0$. If $T^\top p_j - p_i \neq 0$, choose

$$x = \lambda(T^\top p_j - p_i).$$

Then

$$\Pi = \lambda\|T^\top p_j - p_i\|^2 - c_T,$$

which is positive for sufficiently large λ. □

For bounded feasible sets $X \subset \mathbb{R}^d$, the existence problem becomes a linear optimization problem:

$$\max_{x\in X} \left\{(T^\top p_j - p_i)^\top x - C(T)\right\} .$$

11 Topological Arbitrage

Arbitrage may be local, removable, or topologically protected. A local discrepancy can disappear under a small perturbation of valuation functions. A topologically protected discrepancy survives all admissible continuous deformations preserving the transformation structure.

Definition 13 (Arbitrage cycle)**.** *An arbitrage cycle is a closed path γ such that*

$$\Omega(\gamma) \neq 0.$$

It is positive if $\Omega(\gamma) > 0$, negative if $\Omega(\gamma) < 0$.

Definition 14 (Arbitrage homology, informal)**.** *Arbitrage homology classifies closed transformation cycles by the equivalence relation*

$$\gamma_1 \sim \gamma_2$$

if $\gamma_1 - \gamma_2$ *bounds a collection of zero-curvature arbitrage cells.*

A full arbitrage homology theory would assign groups

$$H_k^{\mathrm{arb}}(\mathcal{A})$$

whose nontrivial elements represent persistent arbitrage structures of dimension k.

12 Category-Theoretic Arbitrage

Let $\mathcal{C}$ be a category of structures and $\mathcal{D}$ a category of valuations. A valuation system is a functor

$$F : \mathcal{C} \to \mathcal{D}.$$

Definition 15 (Functorial arbitrage)**.** *Given two functors* $F, G : \mathcal{C} \to \mathcal{D}$*, a morphism* $f : X \to Y$ *in* $\mathcal{C}$ *generates functorial arbitrage if there exists a valuation comparison map* Δ *such that*

$$\Delta(G(f), F(f)) > C(f).$$

This captures situations where two structures appear equivalent under one functor but inequivalent under another. For example, two statistical models may be observationally equivalent but causally distinct; two political institutions may be constitutionally similar but strategically different; two military positions may be geographically close but logistically unequal.

13 Economics and Finance

13.1 General equilibrium interpretation

Let X be the set of allocations, prices, and institutional states:

$$x = (a, p, \iota),$$

where a is an allocation, p is a price vector, and ι is an institutional configuration.

Different actors i have valuations

$$V_i(x) = U_i(a_i) - R_i(p, \iota),$$

where U_i is utility and R_i is regulatory, informational, or liquidity cost.

An economic arbitrage is a transformation T such that

$$U_j((Tx)_j) - R_j(Tp, T\iota) - U_i(x_i) + R_i(p, \iota) > C(T).$$

13.2 Financial no-arbitrage

In a one-period financial market with price vector $S_0 \in \mathbb{R}^d$ and future payoff vector $S_1(\omega)$, a portfolio $\theta \in \mathbb{R}^d$ is a classical arbitrage if

$$\theta^\top S_0 \le 0, \qquad \theta^\top S_1(\omega) \ge 0 \ \forall \omega,$$

and

$$\theta^\top S_1(\omega) > 0$$

for at least one ω.

Arbitrage Geometry embeds this in a valuation system:

$$V_0(\theta) = -\theta^\top S_0, \qquad V_1^\omega(\theta) = \theta^\top S_1(\omega).$$

The arbitrage condition becomes

$$V_0(\theta) \geq 0, \qquad V_1^\omega(\theta) \geq 0,$$

with strict positivity somewhere.

13.3 Liquidity arbitrage

Let $L_i(x)$ be liquidity value in market i. If a position can be transformed from market i to market j, liquidity arbitrage exists when

$$L_j(Tx) - L_i(x) - C_{\text{transfer}}(T) > 0.$$

14 Political Science and Institutional Arbitrage

Political systems contain multiple valuations: legality, legitimacy, coercive capacity, coalition stability, bureaucratic capacity, and narrative power.

Let

$$V_i(x) = \alpha_i L(x) + \beta_i B(x) + \gamma_i K(x) + \delta_i N(x),$$

where

$$L = \text{legality}, B = \text{bureaucratic capacity}, K = \text{coercive capacity}, N = \text{narrative legitimacy}.$$

A political actor has institutional arbitrage if a transformation T satisfies

$$V_i(Tx) - V_i(x) > C_{\text{constitutional}}(T) + C_{\text{coalitional}}(T) + C_{\text{reputational}}(T).$$

Example 1 (Constitutional arbitrage). *A political actor may exploit a gap between formal constitutional rules and informal enforcement norms. The arbitrage cell is the local region where legality and legitimacy diverge:*

$$A_{\text{legal,legit}}(x) = V_{\text{legal}}(x) - V_{\text{legit}}(x).$$

15 Geopolitics and International Relations

Let X be the international state space. A state $x \in X$ includes territorial control, alliance commitments, trade networks, military deployments, industrial capacity, technological capability, and diplomatic reputation.

For actor i, define

$$V_i(x) = \alpha_i M_i(x) + \beta_i E_i(x) + \gamma_i D_i(x) + \delta_i T_i(x) + \eta_i R_i(x),$$

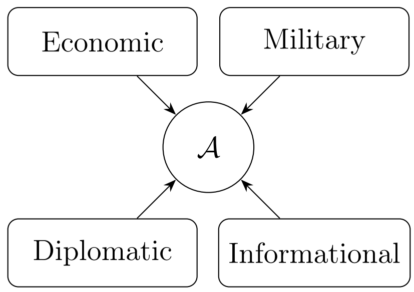
where

$$M_i = \text{military position}, \quad E_i = \text{economic strength}, \quad D_i = \text{diplomatic alignment},$$

$$T_i = \text{technological capacity}, \quad R_i = \text{reputation}.$$

A geopolitical arbitrage exists when an action T changes the international configuration in favor of actor i at lower cost than the induced strategic gain:

$$V_i(Tx) - V_i(x) > C_{\text{military}}(T) + C_{\text{economic}}(T) + C_{\text{diplomatic}}(T) + C_{\text{sanction}}(T).$$



Geopolitical arbitrage cell

16 Warfare Economics and Warfare Arbitrage

War is a transformation system under extreme cost. Let x represent the theater state:

$$x = (q, \ell, m, s, r, \tau),$$

where q is force quality, ℓ is logistics, m is morale, s is sensor coverage, r is resource stock, and τ is terrain.

For commander i, define combat valuation

$$V_i(x) = \alpha_i q_i + \beta_i \ell_i + \gamma_i m_i + \delta_i s_i + \eta_i r_i + \zeta_i \tau_i.$$

A military operation T is profitable in the arbitrage-geometric sense if

$$V_i(Tx) - V_i(x) > C_{\text{casualty}}(T) + C_{\text{logistics}}(T) + C_{\text{political}}(T) + C_{\text{time}}(T).$$

16.1 Asymmetric warfare

In asymmetric warfare, the weaker actor seeks transformations T for which the stronger actor's cost function is much larger:

$$C_{\text{strong}}(T) \gg C_{\text{weak}}(T).$$

Thus a weak actor can obtain positive relative arbitrage even when absolute resources are inferior:

$$\Pi_{\text{weak}}(T, x) = V_{\text{weak}}(Tx) - V_{\text{weak}}(x) - C_{\text{weak}}(T) > 0.$$

16.2 Attrition curvature

Let γ be a campaign cycle. Define attrition curvature as

$$\Omega_{\text{attr}}(\gamma) = \sum_{k=1}^{m} \left[\Delta V_{\text{combat}}(T_k, x_{k-1}) - C_{\text{attrition}}(T_k)\right].$$

A positive attrition curvature means that a sequence of operations compounds advantage. A negative attrition curvature means that tactical actions consume strategic value.

17 Statistics

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be a statistical model class. Let L_i be loss function i, and define

$$V_i(P_\theta) = -\operatorname{E}_{Y \sim P^*}[L_i(Y, P_\theta)].$$

A model transformation $T : \mathcal{P} \to \mathcal{P}$ creates statistical arbitrage if

$$\operatorname{E}[L_i(Y, P_\theta)] - \operatorname{E}[L_j(Y, TP_\theta)] > C(T).$$

17.1 Misspecification arbitrage

Suppose $P^* \notin \mathcal{P}$. The misspecification gap is

$$M(\theta) = \mathrm{KL}(P^* \| P_\theta).$$

A transformation T produces misspecification arbitrage if

$$M(\theta) - M(T\theta) > C(T).$$

17.2 Estimator arbitrage

Let $\hat{\theta}_1, \hat{\theta}_2$ be two estimators. Define

$$V_i(\hat{\theta}) = -\mathrm{MSE}_i(\hat{\theta}).$$

Estimator arbitrage exists if

$$\mathrm{MSE}_i(\hat{\theta}_1) - \mathrm{MSE}_j(\hat{\theta}_2) > C(\hat{\theta}_1 \to \hat{\theta}_2).$$

18 Data Science and Machine Learning

Let X be a representation space, h_θ a predictive model, and $R(h_\theta)$ the population risk:

$$R(h_\theta) = \mathrm{E}[L(Y, h_\theta(X))].$$

A representation map $T : X \to Z$ generates representation arbitrage if

$$R(h_\theta \circ T) - R(h_\phi) < -C(T),$$

or equivalently,

$$R(h_\theta \circ T) - R(h_\phi) + C(T) < 0.$$

18.1 Generalization arbitrage

Let empirical risk be

$$\hat{R}_n(h) = \frac{1}{n} \sum_{k=1}^{n} L(y_k, h(x_k)).$$

Define generalization gap

$$G(h) = R(h) - \hat{R}_n(h).$$

A learning algorithm exploits generalization arbitrage if it finds h such that

$$\hat{R}_n(h) \text{ is low}, \qquad G(h) \text{ is controlled}, \qquad C(h) \text{ is acceptable}.$$

18.2 AI alignment arbitrage

Let

$$V_{\text{train}}(h), \quad V_{\text{deploy}}(h), \quad V_{\text{human}}(h)$$

represent training objective value, deployment performance, and human-aligned value. Alignment arbitrage occurs when a model improves one valuation while degrading another:

$$V_{\text{train}}(h) - V_{\text{human}}(h) > C_{\text{detect}}(h).$$

This captures reward hacking, proxy gaming, benchmark overfitting, and objective misspecification.

19 Algorithmic Framework

19.1 Detection problem

Given finite X, finite $\mathcal{T}$, and finite $\mathcal{V}$, detect whether positive arbitrage exists.

$$\max_{x,T,i,j} \Pi(T, x; i, j) > 0.$$

19.2 Algorithm 1: Exhaustive arbitrage detection

Input	$X, \mathcal{T}, \mathcal{V}, C$
Output	All positive arbitrage tuples
	Initialize $\mathcal{B} \leftarrow \varnothing$.
	For each $x \in X$:
	For each $T \in \mathcal{T}$ such that Tx is defined :
	For each $(i, j) \in I \times I$:
	$\Pi \leftarrow V_j(Tx) - V_i(x) - C(T)$.
	If $\Pi > 0$, add (T, x, i, j, Π) to $\mathcal{B}$.
	Return $\mathcal{B}$.

19.3 Algorithm 2: Curvature-based cycle detection

Input	Transformation graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$
Output	Positive arbitrage cycles
	Assign edge weights $w(e) = V_j(y) - V_i(x) - C(T)$.
	Enumerate simple directed cycles γ.
	For each cycle γ :
	$\Omega(\gamma) \leftarrow \sum_{e \in \gamma} w(e)$.
	If $\Omega(\gamma) > 0$, record γ.
	Return all recorded cycles.

`The following space was deliberately left blank.`

20 Comparative Domain Table

Domain	State x	Valuations V_i	Arbitrage transformation T
Finance	Portfolio, venue, time	Price, payoff, liquidity, risk	Trade, hedge, rebalance
Economics	Allocation, price, institution	Utility, welfare, rent, scarcity	Exchange, regulation, production
Political science	Coalition, institution, legitimacy	Power, legality, legitimacy	Reform, agenda control, coalition shift
Geopolitics	Alliance, territory, capability	Security, influence, reputation	Treaty, sanction, deployment
Warfare	Theater state, logistics, morale	Combat power, attrition, time	Maneuver, deception, strike
Statistics	Model, estimator, sample	Risk, likelihood, loss	Reparameterization, regularization
Machine learning	Data, representation, model	Accuracy, loss, latency, alignment	Feature map, architecture shift
Information theory	Message, code, channel	Entropy, compression, redundancy	Encoding, decoding, compression

21 Classification of Arbitrage Types

Type	Mathematical signature	Interpretation
Local arbitrage	$\exists x, T, i, j : \Pi > 0$	Immediate exploitable discrepancy
Global arbitrage	$\sup \Pi > 0$ over X	System-wide mispricing
Curvature arbitrage	$\Omega(\gamma) > 0$	Net gain around a loop
Topological arbitrage	$[\gamma] \neq 0 \in H_1^{\mathrm{arb}}$	Persistent structural discrepancy
Statistical arbitrage	$\Delta R > C(T)$	Predictive gain exceeds complexity
Computational arbitrage	Δruntime $> C(T)$	Representation advantage
Political arbitrage	Δpower $> C(T)$	Institutional or coalition leverage
Warfare arbitrage	Δcombat value $> C(T)$	Strategic-operational advantage

22 Research Program

Problem 1 (Classification). *Classify all finite arbitrage spaces $\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C)$ satisfying global no-arbitrage.*

Problem 2 (Curvature). *Determine conditions under which every positive arbitrage opportunity lies on a positive-curvature cycle.*

Problem 3 (Embedding). *Given an arbitrage space with positive curvature, determine whether there exists an embedding*

$$\phi : \mathcal{A} \hookrightarrow \mathcal{A}'$$

into a higher-dimensional arbitrage space such that $\mathcal{A}'$ is flat.

Problem 4 (Homology). *Construct groups $H_k^{\mathrm{arb}}(\mathcal{A})$ classifying persistent arbitrage structures.*

Problem 5 (Statistical learning). *Given a hypothesis class $\mathcal{H}$, characterize transformations T that create genuine generalization arbitrage rather than overfitting.*

Problem 6 (Strategic stability). *In political and military systems, characterize the institutional constraints required to suppress destructive arbitrage while preserving productive arbitrage.*

23 Discussion

Arbitrage Geometry is not merely a generalization of financial arbitrage. It is a proposed mathematical theory of how value differences, transformation systems, and costs interact. Its central idea is that many systems become intelligible only when one studies misalignment as a structural object.

In classical geometry, curvature measures deviation from flatness. In Arbitrage Geometry, curvature measures deviation from valuation consistency. In classical economics, arbitrage is eliminated in equilibrium. In Arbitrage Geometry, arbitrage may be local, global, topological, statistical, computational, institutional, or strategic. In machine learning, arbitrage appears when representation changes create performance gains exceeding complexity costs. In geopolitics and warfare, arbitrage appears when an actor converts asymmetry into advantage.

The theory therefore suggests a unified language for several phenomena usually studied separately: mispricing, asymmetry, misspecification, strategic surprise, representation gain, institutional loopholes.

24 Conclusion

This paper proposed Arbitrage Geometry as a new mathematical branch. Its primitive object is the arbitrage space

$$\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C),$$

and its central functional is

$$\Pi(T, x; i, j) = V_j(Tx) - V_i(x) - C(T).$$

The theory studies when this quantity is positive, how it accumulates around loops, when it can be represented by a global potential, and when it becomes a persistent structural feature of the system.

The No-Arbitrage Flatness Theorem shows that zero arbitrage curvature on every closed loop is equivalent to the existence of a global potential representation. This result provides the conceptual bridge between arbitrage, geometry, topology, economics, statistics, computation, and strategy.

Arbitrage Geometry should exist because modern systems are not governed by a single metric of value. They are governed by multiple, conflicting, partially convertible metrics. The mathematical study of such structured misalignment is therefore not peripheral; it is central.

References

[1] K. J. Arrow and G. Debreu, *Existence of an Equilibrium for a Competitive Economy*, Econometrica, 1954.

[2] F. Black and M. Scholes, *The Pricing of Options and Corporate Liabilities*, Journal of Political Economy, 1973.

[3] F. Delbaen and W. Schachermayer, *A General Version of the Fundamental Theorem of Asset Pricing*, Mathematische Annalen, 1994.

[4] J. M. Harrison and D. M. Kreps, *Martingales and Arbitrage in Multiperiod Securities Markets*, Journal of Economic Theory, 1979.

[5] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.

[6] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal, 1948.

[7] A. N. Kolmogorov, *Foundations of the Theory of Probability*, Chelsea Publishing, 1956.

[8] J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton University Press, 1944.

[9] J. Nash, *Non-Cooperative Games*, Annals of Mathematics, 1951.

[10] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, 1960.

[11] C. von Clausewitz, *On War*, 1832.

[12] A. H. Jomini, *The Art of War*, 1838.

[13] J. Pearl, *Causality: Models, Reasoning, and Inference*, Cambridge University Press, 2009.

[14] V. N. Vapnik, *Statistical Learning Theory*, Wiley, 1998.

[15] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

[16] D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge University Press, 2003.

[17] S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1971.

[18] M. Gromov, *Metric Structures for Riemannian and Non-Riemannian Spaces*, Birkhauser, 1999.

Glossary

Admissible transformation
A permitted map $T : X \to X$ that changes the state of the system.

Arbitrage cell
A local region of a state space containing positive arbitrage.

Arbitrage curvature
The net arbitrage accumulated around a closed transformation path.

Arbitrage functional
The quantity $\Pi(T, x; i, j) = V_j(Tx) - V_i(x) - C(T)$.

Arbitrage gradient
The local direction of increasing valuation discrepancy.

Arbitrage homology
A proposed invariant classifying persistent arbitrage cycles.

Arbitrage space
A quadruple $(X, \mathcal{T}, \mathcal{V}, C)$ consisting of states, transformations, valuations, and costs.

Arbitrage tensor
A differential object measuring the sensitivity of multiple valuations to state changes.

Computational arbitrage
Gain from changing representation or algorithm beyond the cost of doing so.

Geopolitical arbitrage
Strategic advantage obtained by exploiting differences among military, economic, diplomatic, and informational valuations.

No-arbitrage equilibrium
A state from which no admissible positive arbitrage transformation exists.

Statistical arbitrage
Predictive or inferential gain exceeding the cost of model transformation.

Topological arbitrage
Arbitrage that persists under admissible continuous deformation.

Warfare arbitrage
Operational or strategic advantage whose value exceeds casualty, logistics, time, and political costs.

The End